DMG80600Y104_04NC

产品特点：

(1) 基于 T5L1 ASIC CPU，支持 DGUSII 人机交互软件平台/标准指令集（TA）。

(2) 10.4 英寸，800*600 分辨率，16.7M 种颜色真彩显示，TN-TFT 液晶显示屏。

(3) 电容触摸屏。

Features:

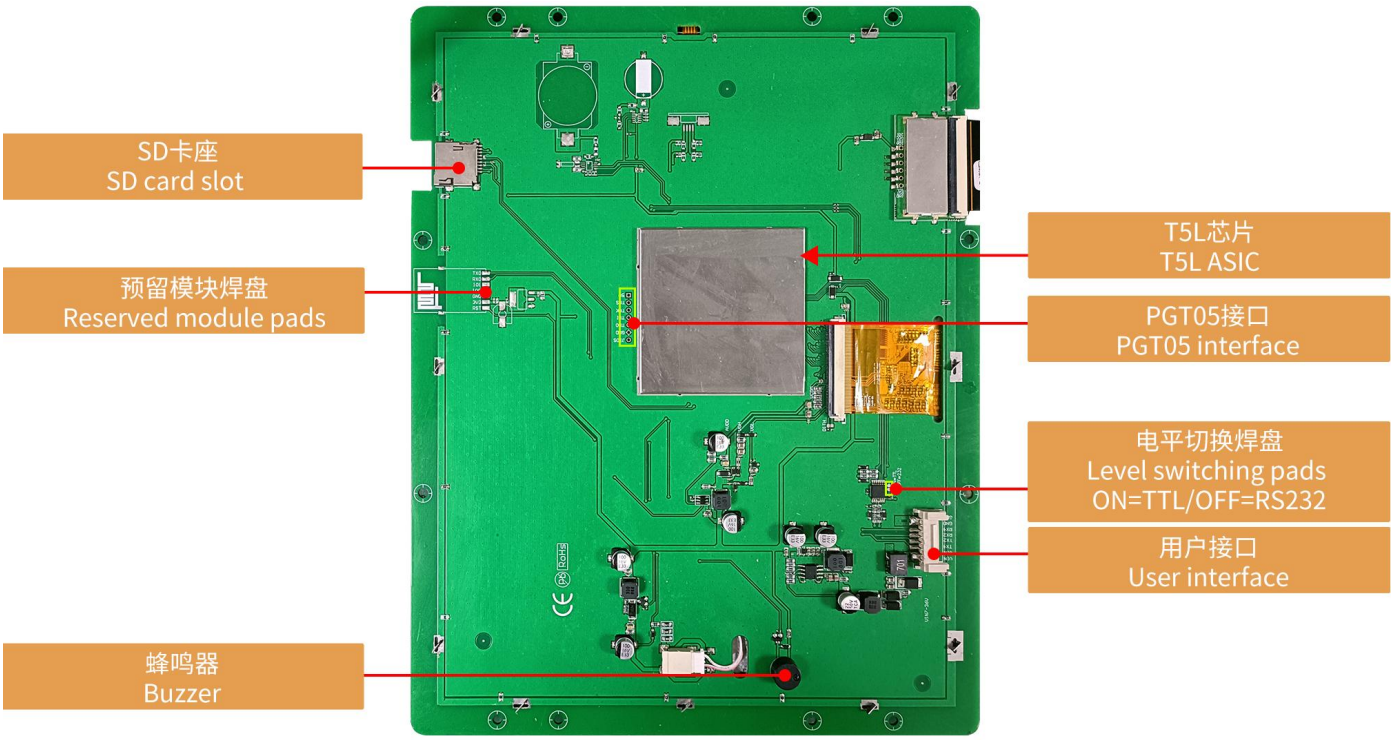
- **Based on the T5L1 ASIC CPU, Support DGUSII human-machine interaction software platform/standard instruction set(TA).**
- **10.4-inch, 800*600 pixels resolution, 16.7M colors true-color display, TN-TFT LCD screen.**
- **Capacitive touch screen.**



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1、硬件及接口 Hardware and interface

1.1 硬件接口图 Hardware interface



硬件接口图
Hardware interface

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1.2 硬件及接口说明 Hardware and interface description

序号 No.	名称 Name	说明 Description
1	T5L1 芯片 T5L1 ASIC	迪文自主研发，2019 年量产；专利加密技术，确保代码和数据安全；功耗低，抗干扰能力强，双面 PCB 设计上轻松通过 EMC/EMI 测试。 DWIN independently developed, mass production in 2019; patented encryption technology ensures code and data security; low power consumption, strong anti-interference capability, easily passes EMC/EMI tests with dual-sided PCB design.
2	用户接口 User interface	用于供电和串口通讯，8Pin_2.0mm 座子。串口下载速率（典型值）：12KByte/s。 8Pin_2.0mm socket for power supply and serial communication. Download rate(typical value): 12KByte/s.
3	Flash	16MBytes NOR Flash，可用于存放字库、图片、音乐等用户 UI 文件，擦写次数>100,000 次。 16MBytes NOR Flash, can be used to store user UI files such as fonts, images, music, etc., with erase/write cycle >100,000 times.
4	扩展 Flash 焊盘 Expand Flash pads	有 3 个扩展位，支持扩展 NOR Flash、NAND Flash。 NOR Flash 最大可拓展到 64Mbytes。 NOR Flash 与 NAND Flash 组合最大可扩展到 48Mbytes（2 个扩展位）+512Mbytes。 Flash 扩展时需焊接译码器、电容等元器件。相关定制可咨询对应业务员（位于屏蔽罩内部） Three expansion slots are available, supporting expansion of NOR Flash and NAND Flash. The maximum expansion for NOR Flash is up to 64Mbytes. When combining NOR Flash and NAND Flash, the maximum expansion is up to 48Mbytes (using two expansion slots) + 512Mbytes. When expanding Flash, components such as decoders and capacitors need to be soldered. Please consult the corresponding salesperson for relevant customization (Located inside the shielding cover)
5	蜂鸣器 Buzzer	3V 无源蜂鸣器。 3V passive buzzer.
6	SD 卡座 SD card slot	支持所有文件（用户 UI 文件、CFG 硬件配置文件、底层内核固件）下载，屏幕显示下载统计信息，下载速率：4Mb/s。下载文件时，SD 卡需格式化为 FAT32 格式，推荐块大小为 4096。 Supports downloading of all files (user UI files, CFG files, underlying kernel firmware), displays download statistics on the screen, download rate: 4 Mb/s. When downloading files, the SD card needs to be formatted in FAT32 format, with a recommended allocation unit size of 4096.
7	预留模块接口 Reserved module interface	支持焊接 WI-FI 模块、USB 下载模块。 WI-FI 模块型号：WI-FI-10；USB 模块型号：HDL702、HDL703。 Supports soldering of WI-FI module and USB download module. WI-FI module model: WI-FI-10; USB module models: HDL702, HDL703.
8	PGT05 接口 PGT05 interface	用于重新烧录 DGUS 底层程序。 Used for reprogramming the underlying DGUS firmware.

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2、规格参数 Specification parameters

2.1 显示参数 Display parameters

显示屏类型 **LCD Type**	TN-TFT 液晶显示屏。 TN-TFT LCD screen.
视角 **Viewing Angle**	普通视角，典型值 70°/70°/50°/70°（L/R/U/D） Normal viewing angle, 70°/70°/50°/70° (L/R/U/D)
分辨率 **Resolution**	800×600（支持 0°/90°/180°/270°显示模式） 800×600 pixels (0°/90°/180°/270°)
AA 区 **Active Area (A.A.)**	211.2mm (W)×158.4mm (H)
VA 区 **View Area (V.A.)**	-
接口 **Interface**	RGB
背光模式 **Backlight Mode**	LED
背光寿命 **Backlight Service Life**	>10000 小时（以最高亮度连续工作，亮度减半时间） >10000 hours (Time of the brightness decaying to 50% on the condition of continuous working with the maximum brightness)
背光亮度 **Brightness**	250nit
背光调节 **Brightness Control**	100 级亮度调节（当亮度调节至最高亮度的 1%~30%时，可能出现闪烁现象，不建议在此范围使用） 0~100 grade (When the brightness is adjusted to 1%~30% of the maximum brightness, flickering may occur and is not recommended to use in this range)
注：长时间固定画面显示可能导致屏幕残影，可以用动态图片切换做屏保解决。 Note: You can use dynamic screen saver wallpapers to avoid afterimages caused by fixed page display for a long time.	

2.2 触摸参数 Touch parameters

触摸屏类型 **Type**	电容式触摸面板。 Capacitive touch panel.
触摸屏结构 **Structure**	G+G 结构，表层为钢化玻璃，玻璃硬度≥6H。 G+G structure with tempered glass surface and hardness ≥ 6H.
透光率 **Light Transmittance**	>85%

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2.3 串口参数 Serial interface parameters

串口模式 Mode	UART2: ON=TTL/CMOS; OFF=RS232 UART4: ON=TTL/CMOS; OFF=RS232 (TA 指令集模式下不可用；DGUS 模式下配置后 OS 后才能使用 Not available in TA instruction set mode. Only available after OS configuration in DGUS)				
串口电平 Voltage Level	测试条件 Test Condition	最小值 Min	典型值 Typ	最大值 Max	单位 Unit
	Output 1, Iout =-4mA	4.78	5.0	-	V
	Output 0, Iout = 4mA	-	-	0.4	V
	Input 1	2.5	5.0	-	V
	Input 0	-	-	1.0	V
串口波特率 Baud Rate	3150~3225600bps，典型值：115200bps。 3150~3225600bps, typical value of 115200bps.				
数据格式 Data Format	UART2: N81 UART4: N81/E81/O81/N82 四种模式可选（OS 配置）4 modes (OS configuration)				
接口排线 Interface Cable	8Pin_2.0mm				

2.4 电气规格 Electrical specifications

额定功率 Rated Power	<5W	
工作电压 Operating Voltage	9~36V，典型值 12V。 9~36V, typical value of 12V.	
工作电流 Operating Current	390mA	VCC=12V，背光亮度最大。 VCC=12V, max backlight.
	125mA	VCC=12V，背光关闭。 VCC=12V, backlight off.
推荐工作电源：12V 1A 的直流稳压电源。 Recommended power supply:12V 1A DC.		

2.5 工作环境 Operating environment

工作温度 Operating Temperature	-10℃~60℃（12V @ 60% RH)
存储温度 Storage Temperature	-20℃~70℃
三防漆工艺 Conformal coating	无 None
工作湿度 Operating Humidity	10%~90%RH，典型值 60%RH。 10%~90%RH, typical value of 60% RH.

2.6 UI 版本 UI

UI 版本 UI Version	DGUS II/TA（出厂预装 DGUS II） DGUS II/TA (DGUS II pre-installed)

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3、可靠性测试 Reliability test

3.1 静电放电测试 Electrostatic discharge test

测试环境温度：25°C，测试环境湿度：50%RH。

Test temperature: 25°C. Test humidity: 50%RH.

试验过程：将产品放置在测试台面的测试工装上（测试工装高度约 15cm），针对智能屏进行接触放电和空气放电测试，实验过程观察屏幕有无死机、黑屏、白屏、花屏、重启等异常现象。性能符合判据 GB/T 17626.2 B 级及以上。

Test process: Place the product on the testing fixture of the test bench (fixture height approximately 15cm), and conduct contact discharge and air discharge tests on the smart screen, during the experimental process, it was observed whether the screen is dead, black, white, splash, or reboot. According to the experiment results, the performance is in line with the criteria GB/T 17626.2 B level and above.

■ Test standard : ☐EN 61000-4-2:2009 ☑IEC 61000-4-2:2008 ☐GB/T 17626.2-2018
☐Other:

Table 1： Electrostatic Discharge Immunity (Air Discharge)

Test Points Locations	Test Levels							
	-2kV	+2kV	-4kV	+4kV	-8kV	+8kV	-15kV	+15kV
屏幕					A	A		
/	/	/	/	/	/	/	/	/
/	/	/	/	/	/	/	/	/

Table 2： Electrostatic Discharge Immunity (Direct Contact)

Test Points Locations	Test Levels							
	-2kV	+2kV	-4kV	+4kV	-6kV	+6kV	-8kV	+8kV
边框					A	A		
/	/	/	/	/	/	/	/	/
/	/	/	/	/	/	/	/	/

性能标准：

A.在制造商、委托方或购买方规定的限值内性能正常；

B.功能或性能暂时丧失或降低，但在骚扰停止后能自行恢复，不需要操作者干预；

C.功能或性能暂时丧失或降低，但需操作者干预才能恢复；

D.因设备硬件或软件损坏，或数据丢失而造成不能恢复的功能丧失或性能降低。

Performance Criterion:

A. Normal performance within limits specified by the manufacturer, requestor or purchaser;

B. Temporary loss of function or degradation of performance which ceases after the disturbance ceases, and from which the equipment under test recovers its normal performance, without operator intervention;

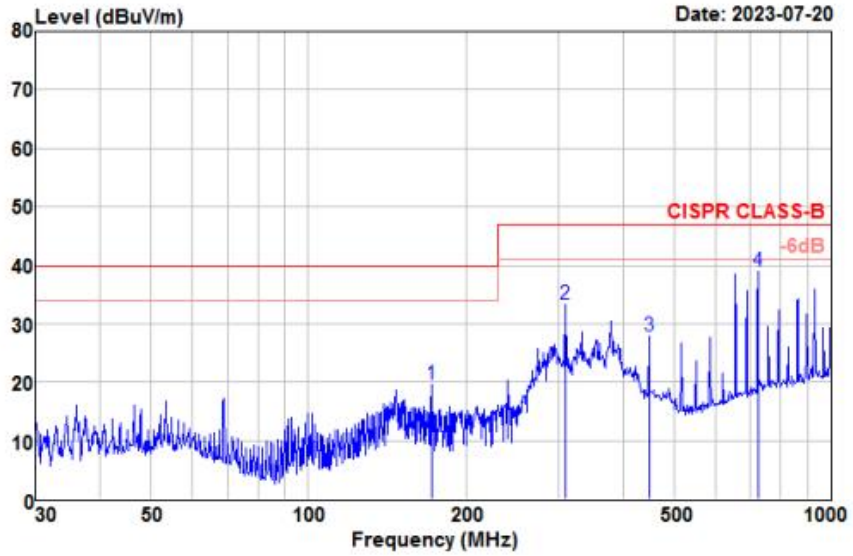
C. Temporary loss of function or degradation of performance, the correction of which requires operator intervention;

D. Loss of function or degradation of performance which is not recoverable, due to damage to hardware or software, or loss of data.

3.2 辐射干扰测试 RE test

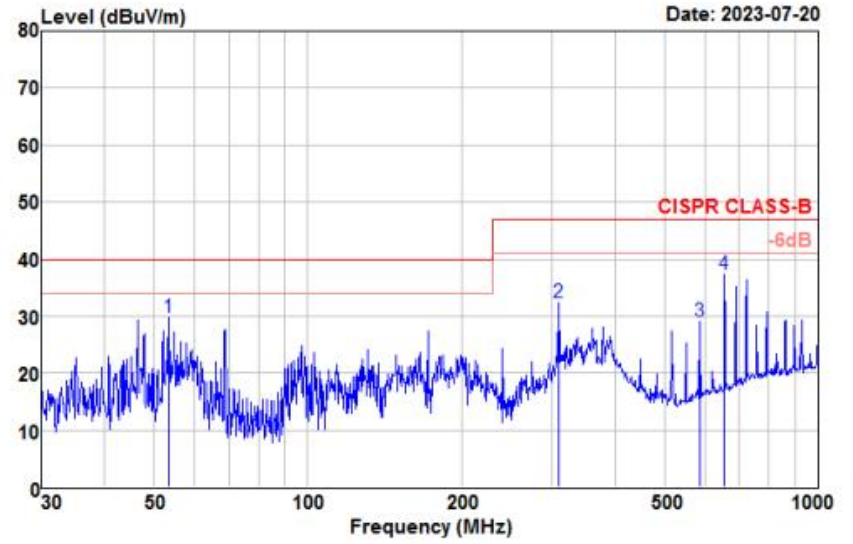
测试项目 Test Item	测试标准 Test Standard	结果 Result
辐射干扰 RE	Class B	正常工作 Normal operation

HORIZONTAL



Freq MHz	Reading level dBuV	Antenna factor dB/m	Cable loss dB	Preamp factor dB	level dBuV/m	Limit level dBuV/m	Over limit dB	Remark
171.995	36.44	13.25	2.53	32.67	19.55	40.00	-20.45	Peak
309.998	49.12	13.38	3.43	32.65	33.28	47.00	-13.72	Peak
447.982	40.27	16.26	4.19	32.79	27.93	47.00	-19.07	Peak
724.261	45.55	20.33	5.42	32.36	38.94	47.00	-8.06	Peak

VERTICAL



Freq MHz	Reading level dBuV	Antenna factor dB/m	Cable loss dB	Preamp factor dB	level dBuV/m	Limit level dBuV/m	Over limit dB	Remark
53.318	45.74	15.26	1.38	32.65	29.73	40.00	-10.27	Peak
309.998	48.16	13.38	3.43	32.65	32.32	47.00	-14.68	Peak
584.790	38.33	18.50	4.94	32.77	29.00	47.00	-18.00	Peak
654.232	45.33	19.38	5.16	32.56	37.31	47.00	-9.69	Peak

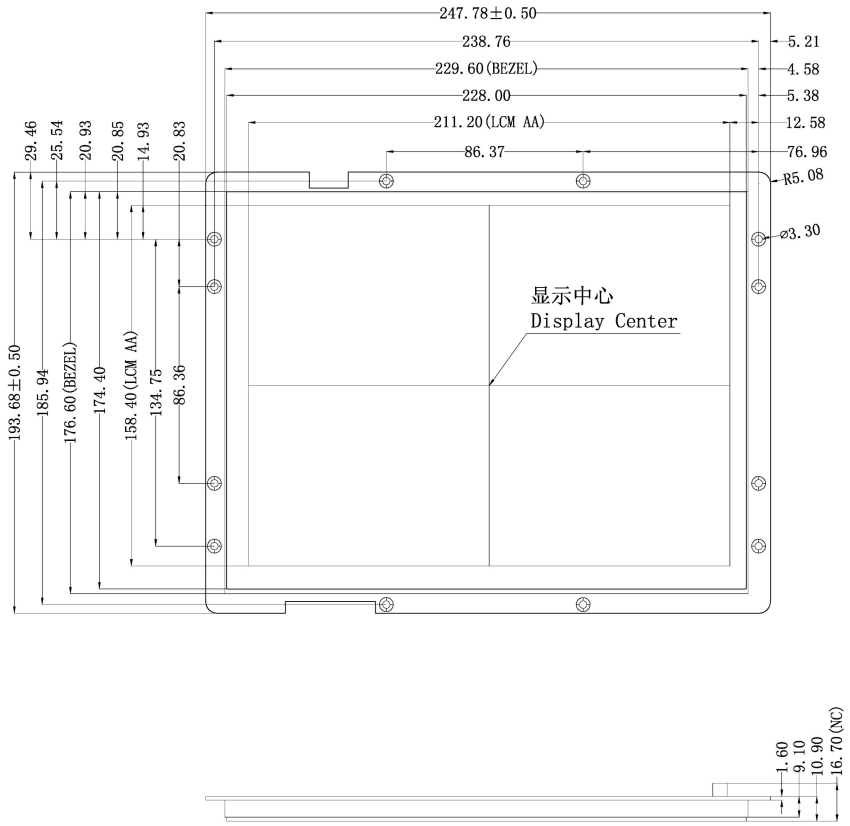
4、包装和物理尺寸 Packaging & dimensions

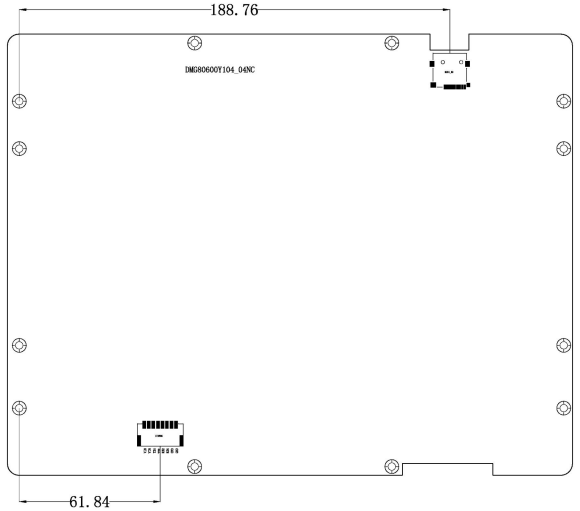
外形尺寸 Form Factor	247.78mm (W)×193.68mm (H)×16.70mm (T)
安装尺寸 Installation Dimensions	开孔尺寸：229.6（+0.3mm）×176.6（+0.3mm） Positioning hole:229.6(+0.3mm)×176.6(+0.3mm)
净重量 Net Weight	725g

包装标准 Packaging Standards

包装箱型号 Model	包装箱尺寸 Dimensions	层数（层） Layer	数量/层（片） Quantity/Layer	总数量（片） Quantity(Pcs)
1 号箱 Carton1：	220mm(L)×160mm(W)×47mm (H)	-	-	-
2 号箱 Carton2：	250mm(L)×200mm(W)×80mm (H)	-	-	-
3 号箱 Carton3：	320mm(L)×270mm(W)×80mm (H)	2	1	2
4 号箱 Carton4：	435mm(L)×335mm(W)×290mm(H)	1	10	10
5 号箱 Carton5：	600mm(L)×430mm(W)×290mm(H)	1	20	20

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定义 Definition	序号 Pin#	类型 Type	说明 Description
VIN	1,2	P	电源输入 Power Input
TX4	3	O	串口4输出 Output
TX2	4	O	串口2输出 Output
RX2	5	I	串口2输入 Input
RX4	6	I	串口4输入 Input
GND	7,8	P	公共接地 GND

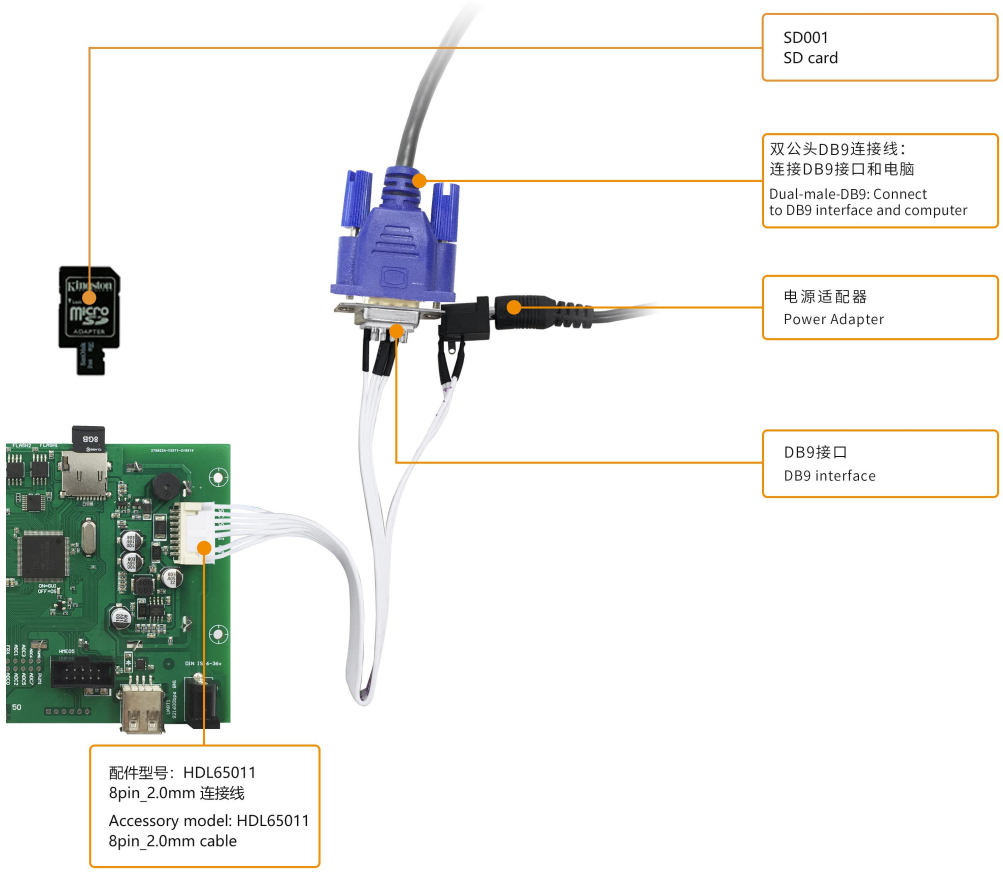
1.尺寸定位基准为定位孔
Location hole is used as position reference.
2.未标注公差为+/-0.3mm
Unmarked Tolerance is +/-0.3mm
注：虚线标注为有效显示区域
Active area is marked in Dash lines

型号 Model	DMG80600Y104-04NC					迪文科技 DWIN Technologies
图纸 Drawing	A 4	绘制 Drawn	G.Y	日期 Date	2024.01.22	
比例 Scale	1:1	审核 Review		日期 Date		
单位 Unit	MM	批准 Approval		日期 Date		

5、调试工具 Debugging tools

建议首次使用迪文智能屏的用户购买标准配件。详细信息可联系客服人员。

It is recommended for new users of DWIN smart LCMs to purchase official accessories. For more details, please refer to customer service center.



迪文科技 技术文

6、T5L 系列芯片特点 T5L series IC features

（1）采用应用最广泛、成熟和稳定的 8051 核，1T（单指令周期）高速工作，最高主频 250MHz。

Mature and stable 8051 core which is the most widely used with the maximum operating frequency of T5L is up to 250MHz, 1T(single instruction cycle)high speed operation.

（2）单独 CPU 核（GUI CPU）运行 DGUS II 系统：

Separate GUI CPU Core running DGUS II System:

- 内置高速显存，2.4GB/S 带宽。

  High-speed display memory, 2.4GB/S bandwidth.

- 2D 硬件加速，JPEG 解压缩速度高达 200fps@1280*800，UI 极其流畅。

  2D hardware acceleration, the decompression speed of JPEG is up to 200fps@1280*800 and the UI with animation and icons as its main feature is extremely cool and smooth.

- JPEG 压缩模式存储图片、图标，大幅度缩小外置存储器到低成本的 16Mbytes SPI Flash。

  Images and icons stored in JPEG format. Adopt Low-cost 16Mbytes SPI Flash.

- 支持电阻或电容触摸屏，灵敏度可以调节，最快 400Hz 触控打点速度。

  Support CTP or RTP with adjustable sensitivity and maximum 400 Hz touch frequency.

- 1 路 15bit 32Ksps PWM 数字功放驱动扬声器，实现高品质语音压缩存储和播放。

  1-way 15bit 32Ksps PWM digital power amplifier driver loudspeaker, save power amplifier cost and achieve high signal-to-noise ratio and sound quality restoration.

- 128Kbytes 变量存储器空间，存储器接口和 OS CPU 核交换数据，应用及其简单。

  128Kbytes variable storage space for exchanging data with OS CPU Core and memory.

- 支持 PC 端组态开发和仿真，支持后台远程升级。

  Support DGUS development and simulation on PC. Support background remote upgrade.

（3）单独 CPU 核（OS CPU）运行用户 8051 代码，应用中省掉用户 CPU：

Separate CPU (OS CPU) core runs user 8051 code or DWIN OS system and user CPU is omitted in practical application:

- 标准 8051 架构和指令集，64Kbytes 代码空间，32Kbytes 片内 RAM。

  Standard 8051 architecture and instruction set, 64Kbytes code space, 32Kbytes on-chip RAM.

- 64bit 整数型数学运算单元（MDU），包括 64bit MAC 和 64bit 除法器。

  64 bit integer mathematical operation unit (MDU), including 64 bit MAC and 64 bit divider.

- 28 个 IO，4 路 UARTs，1 路 CAN 接口，最多 8 路 12bit A/D，2 路 16bit 分辨率可调的 PWM。

  28 IOs, 4-channel UARTs, 1-channel CAN, up to 8-channel 12-bit A/Ds and 2-channle 16-bit PWM of adjustable resolution.

- 支持 IAP 在线仿真和调试，断点数量无限制。

  Support IAP on-line simulation and debugging with unlimited number of breakpoints.

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- 可以透过 DGUS 系统在线升级代码。

  Upgrade code online through DGUS system.

（4）1Mbytes 片内 Flash，迪文专利加密技术，确保代码和数据安全，杜绝山寨和克隆。

1Mbytes on-chip Flash with DWIN patent encryption technology ensure code and data security.

（5）-40℃~+85℃工作温度范围（可定制 -55℃~105℃工作温度范围 IC）。

Operating temperature ranges from -40℃ to +85℃(IC operating temperature customizable from -55℃ to 105℃).

迪文欢迎广大用户基于 T5L 自主设计客制化产品。

DWIN encourages users to design your own customized product based on T5L.

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7、修订记录 Revision records

版本 Rev	日期 Revise Date	描述 Content	编辑人 Editor
00	2022-10-25	首次发布 First Edition	郑运佳
01	2022-11-14	修改尺寸说明 Modify dimension description	郑运佳
02	2022-11-24	出厂预装 DGUS II Factory pre installed DGUS II	郑运佳
03	2024-01-25	更新 CAD 图；EMC 硬件升级，库存旧板自然消耗 Update CAD drawing;EMC hardware upgrade, natural consumption of old inventory boards	赵方维
04	2024-08-30	更新供电电压 Modify Power Voltage	赵方维

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迪文开发者论坛 DWIN Developer Forum: http://inforum.dwin.com.cn:20080/forum.php

感谢大家一直以来对迪文的支持，您的支持是我们进步的动力！

Thank you all for continuous support of DWIN, and your approval is the driving force of our progress!

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重要声明

Important Disclaimer

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对于客户在其产品中采用迪文产品可能引发的任何后果，迪文不承担连带责任。特别地，对于可能导致的重大财产损失、环境危害、人身伤害甚至死亡等风险，尤其是在军事、易燃易爆、生命医疗设备等高风险应用领域，客户应自行评估风险，并采取相应的预防和保护措施，迪文不承担相关责任。

DWIN reserves the right to make any changes to product designs without prior notice.

Customers should ensure strictly adhering to all the relevant standards and requirements during the product application process, including but not limited to functional safety, information security, and regulatory provisions.

DWIN shall not bear any joint and several liability for any consequences that may arise from customers' adoption of DWIN products. In particular, for risks that may lead to significant property losses, environmental hazards, personal injury, or even death, especially in high-risk application areas such as military applications, flammable and explosive places, and life-saving medical equipment, customers should independently assess the risks and take corresponding preventive and protective measures. DWIN shall not bear any relevant responsibility.

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